Exp No:12 Experiment to detect outliers in a given data set.

AIM:

To detect outliers in a given data set.

CODE:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Load CSV file
df = pd.read_csv("experiment12_outliers.csv")

# Convert column to NumPy array
array = df["Value"].values

print("Array:")
print(array)

# Mean
print("\nMean:")
print(array.mean())

# Percentiles
print("\n25th Percentile (Q1):")
print(np.percentile(array, 25))

print("\n50th Percentile (Median):")
print(np.percentile(array, 50))

print("\n75th Percentile (Q3):")
print(np.percentile(array, 75))

print("\n100th Percentile:")
print(np.percentile(array, 100))

# Outlier detection using IQR
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
```

```
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)

print("\nLower Range:", lr)
print("Upper Range:", ur)

# Detect outliers
outliers = array[(array < lr) | (array > ur)]

print("\nOutliers:")
print(outliers)

# Remove outliers
new_array = array[(array > lr) & (array < ur)]

print("\nArray after removing outliers:")
print(new_array)

# Distribution plot
sns.displot(array)
plt.title("Distribution of Data")
plt.show()
```

OUTPUT:

```
            Upload widget is only available when the cell has been executed in the current browser
session. Please rerun this cell to enable.
Saving experiment12_outliers.csv to experiment12_outliers.csv
Array:
[27 50 44  6 58 61 23 86 67 20 75  7 79 61 90 54]

Mean:
50.5

25th Percentile (Q1):
26.0

50th Percentile (Median):
56.0

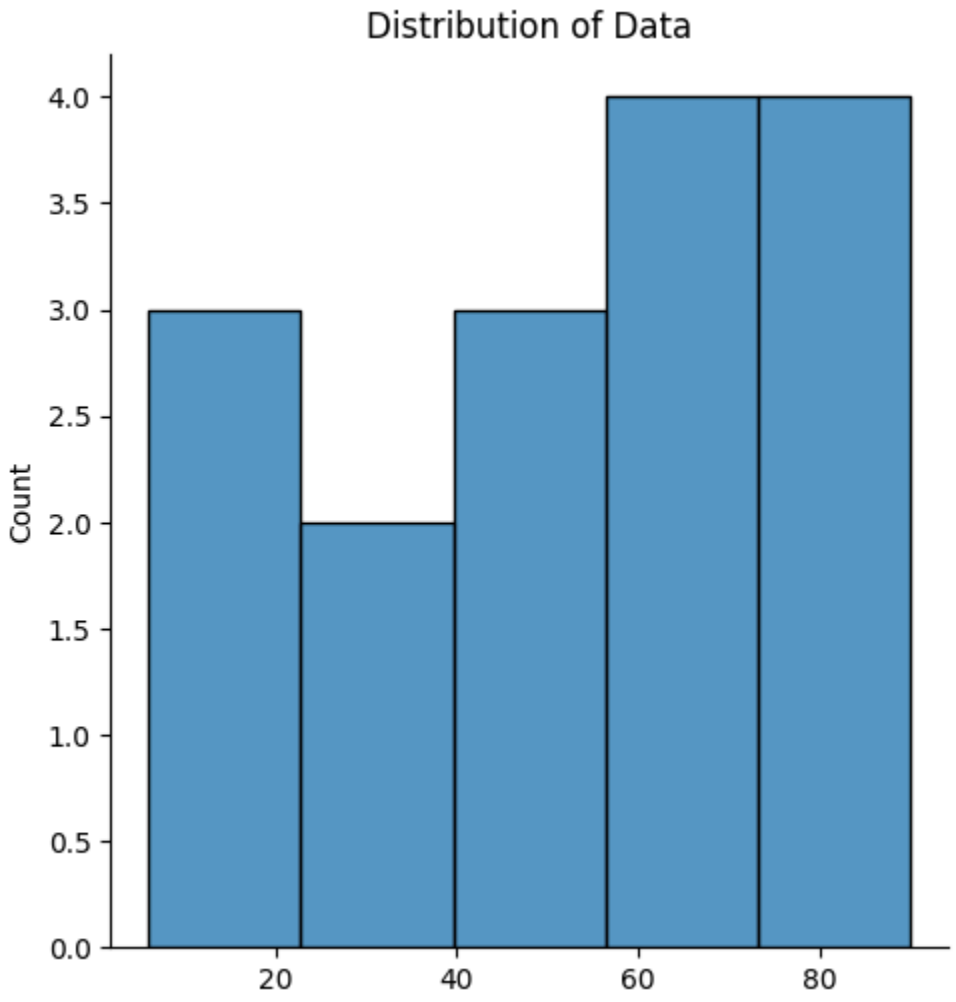
75th Percentile (Q3):
69.0
```

100th Percentile:
90.0

Lower Range: -38.5
Upper Range: 133.5

Outliers:
[]

Array after removing outliers:
[27 50 44 6 58 61 23 86 67 20 75 7 79 61 90 54]



RESULT:

Thus,the given data was analyzed successfully and required output was obtained.